QUESTIONS FOR CHAPTER 8 PROJECT SCHEDULING

1. One of the obvious purposes of project planning and scheduling is to complete a project **on time**. Identify other **benefits** that can be derived from good project planning. Also **describe typical problems** that can be prevented by good planning and scheduling.

   项目规划和排程的一个明显目的是确保项目**按时**完成。请识别良好项目规划所带来的其他**好处**，并**描述**可以通过良好规划和排程**预防的典型问题**。

(1) Benefits of Planning:

- **Centralized tool** for **communication** that coordinates the work of all parties.
- Provides a **benchmark** of **work activities** that can be used to **measure progress**.
- Keeps the work **flowing in a continuous manner** so everyone is **better informed**.
- **Forces project personnel** to identify, in advance, the work that **must be performed**.
- Provides **accountability of people**, with **defined responsibility** and **authority**.
- **Organizes work**, leading to **improved quality** of the project.
- Provides a **framework** within which to **make decisions** with **good information**.

规划的好处：

- 作为一个**集中的沟通工具**，协调各方的工作。
- 提供一个**工作活动的基准**，可以用来**衡量进度**。
- 保持**工作连续进行**，确保所有人**信息更充分**。
- **促使项目人员**提前识别必须执行的工作。
- 提供**人员的责任**，并明确**职责和权力**。
- **组织工作**，从而**提高项目质量**。
- 提供一个**框架**，使得可以在**良好信息**的基础上做出决策。

简化版：

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Benefits of Planning:

- **Communication Tool**: Centralized communication, work coordination.
- **Progress Benchmark**: Measures progress.
- **Continuity**: Keeps information flowing.
- **Advance Identification**: Identifies necessary tasks.
- **Clear Accountability**: Defines roles and authority.
- **Quality Improvement**: Organizes work, enhances quality.
- **Decision Framework**: Supports decision-making with information.

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规划的好处：

- **沟通工具**：集成沟通，协调工作。
- **进度基准**：衡量进度。
- **持续性**：保持信息畅通。
- **提前识别**：确认必做工作。
- **责任清晰**：明确职责和权限。
- **质量提升**：组织工作，改善质量。
- **决策框架**：信息支持决策。

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(2) Problems Prevented:

- **Reduces the number of changes** in a project and the amount of **rework**.
- Helps in **preventing interruptions** and **delays** in work.
- **Reduces confusion** and **misunderstandings** between people on the project.
- Can be used to **reduce and manage legal disputes** that may arise in a project.
- Helps in **preventing low worker morale** and **declining productivity**.
- Helps to **prevent duplication** of work or **overlaps** in work.

预防的问题：

- **减少**项目中的**变更数量**和**返工**的量。
- 帮助**防止工作中的中断和延误**。
- **减少项目人员之间的困惑和误解**。
- 可以用来**减少和管理**项目中可能产生的**法律纠纷**。
- 帮助**防止员工士气低落**和**生产力下降**。
- 帮助**防止工作重复**或**工作重叠**。

简化版：

Problems Prevented:	**预防的问题：**
• **Fewer Changes**: Reduces rework. • **Prevents Interruptions**: Avoids delays. • **Less Misunderstanding**: Smoother communication. • **Reduces Disputes**: Lowers legal risks. • **Boosts Morale**: Prevents productivity decline. • **Prevents Duplication**: Avoids overlap in work.	• **减少变更**：减少返工。 • **防止中断**：避免延误。 • **减少误解**：沟通更顺畅。 • **降低纠纷**：减少法律风险。 • **提高士气**：避免生产力下降。 • **避免重复**：防止工作重叠。

2. A list of **activities** that are required to complete a project is shown below. Draw a **CPM precedence diagram** for the project and calculate the **project schedule**. Provide the schedule in a **tabular form**, showing **early/late starts** and **finishes** as well as the **total** and **free float** for each activity. Denote the **critical path** on the network.

   以下列出了完成项目所需的**活动**列表。请为项目绘制 **CPM（关键路径法）优先图**并计算**项目进度**。以**表格形式**提供进度信息，显示每项活动的**最早/最晚开始和结束时间**，以及**总浮动**和**自由浮动**。在网络图上标注**关键路径**。

Activity	Duration	Cost	Preceded by	Followed by
A	2 days	$500	None	B, C, D
B	3 days	$900	A	E
C	4 days	$1,600	A	F
D	5 days	$500	A	G
E	7 days	$1,400	B	H
F	7 days	$1,500	C	I, L
G	8 days	$2.400	D	J, K
H	4 days	$800	E	L
I	2 days	$1,000	F	N
J	12 days	$3,600	G	M, O
K	5 days	$2,000	G	P
L	6 days	$1,200	F, H	Q
M	2 days	$900	J	N
N	2 days	$700	I, M	S
O	6 days	$1,800	J	R, T
P	4 days	$1,200	K	T
Q	4 days	$2,000	L	U
R	4 days	$1,600	O	S
S	2 days	$1,400	N, R	V
T	9 days	$1,800	O, P	V
U	2 days	$1,200	Q	V
V	3 days	$300	S, T, U	None

3. Perform a **cost analysis** based on an **early start** and **late start schedule** for the project in Question 2. Plot the **S-curve** for the early and late start costs. Show **costs** on the **left-hand ordinate** and **percent costs** on the **right-hand**. Along the **abscissa**, show the **time in working days** and in **percentage of project duration**.

   根据问题 2 中的项目，基于**最早开始时间**和**最晚开始时间**执行**成本分析**。绘制 **S 曲线**，显示最早和最晚开始时间的成本。在**左侧纵轴**标注**成本**，在**右侧纵轴**标注**百分比成本**，在**横轴**上显示**工作日**和**项目持续时间的百分比**。

提示：

问题 2 要求绘制 CPM 图表、计算最早/最晚的开始和结束时间、总浮动和自由浮动，并标注关键路径。
问题 3 则在问题 2 的基础上增加了**成本分析**和 **S 曲线的绘制**，针对最早和最晚开始时间进行成本对比。
问题 6 和问题 8 要求更进一步，除了问题 2 和 3 的内容外，还包括：

- **目标进度**（target schedule）：这是除了早开始和晚开始之外的一个额外进度基准，通常用于计划和实际情况的对比。
- **每日成本分布表格**：需要详细计算并展示每天的成本分布，而不仅是绘制总成本的 S 曲线。
- **三条 S 曲线**：早开始、晚开始和目标进度三种不同计划的成本曲线。

4. Describe **advantages** and **disadvantages** of each of the two basic techniques of **project scheduling**: **bar charts** and **CPM networks**. Include **situations** most appropriate for each of the two techniques.

描述两种基本的**项目排程技术**（**条形图**和 **CPM 网络**）的**优缺点**。包括每种技术最合适的**应用场景**。

(1.1) Advantages of Bar Charts:

- **Simple format** and **easy to interpret** by all project participants, both **technical and non-technical**.
- Effective **planning tool** for projects that do not have extensive **interdependency of work**.
- **Preferred** by many project personnel for planning **short-term work** and/or **simple projects**.
- **Long-established** record for planning and often **preferred for visual displays**.

甘特图的优点：

- **格式简单**，所有项目参与者（**技术和非技术人员**）都**易于理解**。
- 对于**没有大量工作相互依赖**的项目来说，是一种**有效的规划工具**。
- **短期工作**和/或**简单项目**的规划中受到许多项目人员的**青睐**。
- 具有**长期记录**，常被**优先用于可视化展示**。

(1.2) Disadvantages of Bar Charts:

- **Difficult to update** for projects that frequently **get off schedule**.
- Do not show **interdependencies of activities**.
- Do not provide **cost distribution** with respect to **time**.
- Cannot **resource load** the project to show the **utilization of resources**.

甘特图的缺点：

- 对于**频繁偏离进度**的项目来说**难以更新**。
- **无法显示活动的相互依赖关系**。
- **无法显示随时间分布的成本**。
- **无法进行资源加载**，以显示**资源的利用情况**。

简化版：

Bar Charts	甘特图（Bar Charts）
Advantages: - Simple and easy to understand; ideal for **short-term work** and **simple projects** - Effective for projects with **low interdependency** - Strong visual display; widely used historically	**优点**： - 简单易懂，适合**短期工作**和**简单项目** - **无大量依赖性**的项目使用效果佳 - 可视化显示，历史应用广泛
Disadvantages: **- Difficult to update** for frequently changing projects **- Does not show activity dependencies** - Lacks **cost distribution** and **resource utilization** display	**缺点**： - **难更新**（对于频繁变动的项目） - **不显示活动依赖关系** - **无成本分布**和**资源利用**显示
Best suited for: Short-term or simple projects with minimal dependencies	**适用场景**：短期或简单项目，无大量依赖关系

(2.1) Advantages of CPM Networks:

- Provides a **logical display** of all work and **sequencing** of work.
- Many **software packages** are available for **automating** the CPM schedule.
- Can **cost load** and **resource load** the schedule to provide **forecasts of project needs**.
- Offers a good **record of project progress** for **documentation purposes**.
- Provides a variety of **reports** that can be prepared for **internal and external distribution**.
- Can **integrate** the fundamental components of a project: **work, cost, and schedule**.

CPM 网络的优点：

- 提供所有工作及其**顺序的逻辑展示**。
- 有许多**软件工具**可用于**自动化 CPM 进度**。
- 可以**加载成本**和**加载资源**，以提供项目需求的**预测**。
- 提供**项目进展的良好记录**，便于**文件存档**。
- 可生成多种**报告**，用于**内部和外部分发**。
- 能够**整合**项目的基本要素：**工作、成本和进度**。

(2.2) Disadvantages of CPM Networks:

- **Requires time** for the **development of the original network** and **time for updating**.
- The **structured start and finish dates** of activities are sometimes **resisted by team members**.
- **Credibility** of the project schedule can be lost when **team members deviate from the plan**.
- Too often it is used to **settle disputes** rather than for **effective planning** of project work.

CPM 网络的缺点：

- **需要时间**来**开发原始网络**并**进行更新**。
- 活动的**固定开始和结束日期**有时会**受到团队成员的抵触**。
- 当**团队成员偏离计划**时，项目进度的**可信度**可能会降低。
- 经常用于**解决争议**，而非用于**有效规划**项目工作。

简化版：

CPM Networks **Advantages**: **- Logical display** of **all tasks and sequencing** - Allows **cost and resource loading** to forecast project needs - Generates **variety of reports**; good for documentation - Integrates **work, cost, and schedule** **Disadvantages**: **- Time-consuming** to develop and update - Team may resist **fixed start/finish dates** - Schedule **credibility drops** when the plan is not followed **Best suited for**: Complex, long-term projects with high task interdependency	**CPM 网络（CPM Networks）** **优点**： - **逻辑展示**，显示**所有工作和顺序** - **成本**和**资源加载**，预测项目需求 - **多种报告**生成，适合记录和文件存档 - 整合**工作、成本、进度** **缺点**： - **开发和更新耗时** - 团队对**固定开始/结束日期**可能有抵触 - 偏离计划会降低**进度可信度** **适用场景**：复杂、长周期项目，有大量工作依赖关系

5. Provide a brief summary of the **responsibilities** of the **owner, designer, and contractor** related to **project planning** and **scheduling**. **Describe how the schedule** of each party should be **related**, including the **advantages and disadvantages** of maintaining **one common schedule** for a project.

简要总结**业主、设计师和承包商**在**项目规划和排程**中的**职责**。**描述**各方的进度**应如何关联**，包括维护**一个统一的项目进度表**的**优缺点**。

(1) Owner's Responsibilities:

- **Set the approved start date** and the **required finish date** for the project.
- **Establish major milestone dates** to guide the **designer's** and **contractor's work**.
- **Identify priorities** in the project to help team members **schedule work**.
- **Communicate "desired dates"** and **"need dates"** to appropriate project parties.

业主的职责：

- **确定**项目的**批准开工日期**和**要求的完成日期**。
- **设定主要里程碑日期**，以指导**设计师**和**承包商**的工作。
- **识别项目中的优先事项**，帮助团队成员**安排工作**。
- 向相关项目方**传达"期望日期"和"需要日期"**。

简化版：

Owner's Responsibilities:	**业主的职责：**
• Set start and finish dates • Establish milestones • Communicate project priorities • Provide "desired" and "need" dates	• 设定开工和完成日期 • 设定里程碑 • 传达项目优先事项 • 提供"期望"和"需要"日期

(2) Designer's Responsibilities:

- **Develop a design schedule** that meets the dates required by the owner.
- **Produce designs and contract documents** for contractors that align with the owner's schedule.
- **Address questions from the contractor** to prevent **delays in the contractor's work**.
- **Assist in procurement** of special long lead-time equipment designed for the project.
- **Provide schedule status reports** showing the project's progress to the owner.

设计师的职责：

- **制定设计进度表**，以满足业主的时间要求。
- 为承包商**提供符合业主进度要求的设计和合同文件**。
- **回答承包商的疑问**，以避免**承包商工作延误**。
- **协助采购**为项目设计的特殊长交货期设备。
- **向业主提供进度状态报告**，显示项目的进展情况。

简化版：

Designer's Responsibilities:	**设计师的职责：**
• Develop design schedule to meet requirements • Provide design and contract documents • Answer contractor's questions • Assist in procuring special equipment • Provide progress reports	• 制定符合要求的设计进度 • 提供设计和合同文件 • 解答承包商疑问 • 协助采购特殊设备 • 提供进度报告

(3) Contractor's Responsibilities:

- **Develop a detailed construction schedule** to meet the requirements of the contract documents.
- **Include suppliers, vendors,** and **subcontractors** in the construction schedule.
- **Procure materials and equipment** to ensure the project schedule is met.
- **Periodically update the schedule** to reflect the current status of the project.
- **Report the status of the project** to the owner in regularly scheduled reports.

承包商的职责：

- **制定详细的施工进度表**，以符合合同文件的要求。
- 在施工进度表中**包含供应商、卖方和分包商**。
- **采购材料和设备**，以确保符合项目进度。
- **定期更新进度表**，反映项目的最新状态。
- 在定期报告中**向业主汇报项目状态**。

简化版：

Contractor's Responsibilities:	**承包商的职责：**
• Create detailed construction schedule • Include suppliers and subcontractors • Procure materials and equipment • Update schedule regularly • Report progress to the owner	• 制定详细施工进度 • 包含供应商和分包商 • 采购材料设备 • 定期更新进度 • 向业主汇报进展

(4) Advantages and Disadvantages of a Common Schedule:

公共进度表的优缺点：

Advantages:

- With **one common schedule, interference of work between parties** is reduced.
- **Eliminates duplication of efforts** among various parties.
- Each party is more **aware of their impact** on one another.
- **Responsibility and commitment to scheduling** is enhanced.

优点：

- 使用**一个公共进度表**，可**减少各方工作之间的干扰**。
- **消除各方工作中的重复**。
- 每一方**更清楚自己对他方的影响**。
- **增强各方对进度安排的责任和承诺**。

Disadvantages:

- **Liability for the party responsible** for maintaining the common schedule.
- **Coordination issues** from obtaining input from three sources for updates.
- **Deviation from the plan** by workers from the three organizations can cause confusion.
- **Timing of updates** may not align with the **reporting periods** of the three parties.

缺点：

- 负责维护公共进度表的**方承担责任风险**。
- **协调更新进度表**时需从三方获取输入，存在问题。
- **三方的员工可能偏离计划**，从而导致混乱。
- **更新的时间**可能与三方的**正常报告周期不匹配**。

简化版：

Advantages and Disadvantages of a Common Schedule:	**统一进度表的优缺点：**
Advantages: • Reduces interference • Eliminates duplicated work • Increases accountability • Enhances coordination	**优点：** • 减少干扰 • 消除重复工作 • 提高责任意识 • 增强协调性
Disadvantages: • Responsibility risk for maintaining party • Coordination issues among three inputs • Deviation from plan can cause confusion • Update timing may not align	**缺点：** • 维护方有责任风险 • 协调三方输入困难 • 三方偏离计划易混乱 • 更新周期可能不一致

6. A **project manager** is usually assigned the responsibility of either managing a **single large project** or **many small projects** at a time. Describe the **different approaches** to **planning** and **scheduling** that must be used by these two types of project managers.

项目经理通常负责管理一个**大型项目**或同时管理**多个小项目**。描述这两类项目经理在**规划**和**排程**上必须使用的不同**方法**。

Solution:

- For many **small projects**, the problem is that the **project manager** can **lose sight** of some projects.
- For multiple small projects, what is needed is **one schedule** for the **project manager**.
- The **project manager's schedule** contains **all projects** that he or she is managing.
- The **linking of small projects** is determined by **resource utilization**.
- **Scheduling multiple small projects** requires scheduling **projects with shared resources**.

- 对于许多**小项目**来说，问题在于**项目经理**可能**忽视**某些项目。
- 对于多个小项目，需要的是一个**项目经理的统一进度表**。
- **项目经理的进度表**包含他或她正在管理的**所有项目**。
- **小项目的关联**由**资源的利用**来决定。
- **安排多个小项目的进度**需要**协调共享资源的项目**。

简化版：

Large Project Management: • **Single-project focus** • **Schedule and resources dedicated to one objective**	**大型项目管理：** • **单一项目集中管理** • **进度和资源专注于一个目标**
Small Project Management: • **Unified schedule**: includes all small projects • **Shared resources**: coordinates resources across projects • **Prevents oversight**: avoids losing track of projects	**小项目管理：** • **统一进度表：**包含所有小项目 • **资源共享：**协调各项目的资源 • **防止忽视：**避免项目失控

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7. Describe the **purpose** and **use** of a **coding system** for activities in a **CPM project schedule**. Include **factors** that should be considered in the **development of the coding system** and describe how the **coding system** can relate to the **work breakdown structure**, **organizational breakdown structure**, and the **cost breakdown structure**.

描述在 **CPM 项目进度表**中**活动编码系统**的**目的**和**用途**。包括编码系统开发中应考虑的**因素**，以及编码系统如何与**工作分解结构**、**组织分解结构**和**成本分解结构**相关联。

Solution:

(1) Purpose and Use of the Coding System:

- The **coding system identifies** the **contents, parts, and subparts** of a project.
- **Provides a method** of **sorting and analyzing** specific parts within the total project.
- **Allows reports** to be tailored to the **specific needs and interests** of project participants.
- **Provides structure** to **evaluate performance** for specific parts of the project.

编码系统的目的和用途：

- **编码系统识别**项目的**内容、部分和子部分**。
- **提供一种方法**，用于在整个项目中**分类和分析**特定部分。
- **允许生成报告**，以满足项目参与者的特定需求和兴趣。
- **提供结构**，用于**评估项目特定部分的绩效**。

简化版：

Purpose and Use of the Coding System:	编码系统的目的和用途：
• **Identify** project contents and parts • **Sort and analyze** project components • **Customize reports** to meet needs • **Structure for performance** evaluation	• **识别**项目内容和部分 • **分类和分析**项目部分 • **定制报告**以满足需求 • **绩效评估**的结构

(2) Factors to Consider in a Coding System:

- **Phases of the project**: such as engineering, procurement, construction.
- **Facilities of the project**: such as grading, drainage, utilities, buildings, process units.
- **Geographic areas** of the project: such as north side, upper level, station number, etc.
- **Technical areas of expertise**: such as civil, electrical, mechanical, structures, etc.
- **Responsible parties**: such as team members, vendors, subcontractors, etc.
- **Unique areas** of the project.

编码系统需要考虑的因素：

- **项目阶段**：如工程、采购、施工。
- **项目设施**：如场地平整、排水、公用设施、建筑物、工艺单元。
- **地理区域**：如北侧、上层、站号等。
- **技术专长领域**：如土木、电气、机械、结构等。
- **责任方**：如团队成员、供应商、分包商等。
- **项目的独特区域**。

简化版：

Factors to Consider in a Coding System:	编码系统的考虑因素：
• **Project Phases**: engineering, procurement, construction • **Project Facilities**: drainage, buildings, utilities • **Geographic Areas**: north side, upper level, etc. • **Technical Fields**: civil, electrical, mechanical • **Responsible Parties**: team members, vendors • **Unique Areas**: project-specific zones	• **项目阶段**：工程、采购、施工 • **项目设施**：排水、建筑、公用设施 • **地理区域**：北侧、上层等 • **技术领域**：土木、电气、机械 • **责任方**：团队成员、供应商 • **独特区域**：项目特定区域

Relating Codes to the WBS, OBS, and CPM:

- **Fields of digits** in the code can be earmarked to **represent each level** of the WBS, OBS, and CPM.
- Therefore, a **single code for a work item** can indicate its **relationship** to the WBS, OBS, and CPM.

编码与 **WBS**、**OBS** 和 **CPM** 的关系：

- **编码中的数字字段**可以用于**表示 WBS、OBS 和 CPM 的各个级别**。
- 因此，**一个工作项的单一编码**可以表明其与 **WBS**、**OBS** 和 **CPM** 的关系。

简化版：

Relation of Coding to WBS, OBS, CBS:	编码与 WBS、OBS、CBS 的关系：
• Numeric fields represent **WBS, OBS, CBS levels** • **Single code** shows **relationship to breakdown structures**	• 数字字段代表 **WBS、OBS、CBS 级别** • **单一编码**显示**各分解结构的关联**

8. The **data** for the **CPM diagram** for a project are shown below. Perform a **cost analysis** for the project and present the **results** in **tabular form** showing the **daily distribution costs** on an **early start, late start**, and **target schedule**. Plot the **three S-curves** on one graph to show the results of your analysis.

以下是项目的 **CPM 图表数据**。对项目进行**成本分析**，并以**表格形式**展示结果，显示**早期启动**、**晚期启动**和**目标进度**的**每日成本分布**。在一个图上绘制三条 **S 曲线**，显示分析结果。

Activity	Duration	Cost	Preceded by	Followed by
A	2 days	$1,200	None	B, C, D
B	3 days	$3,300	A	E
C	8 days	$12,000	A	F
D	6 days	$18,000	A	F, G, H
E	8 days	$8,000	B	J
F	9 days	$27,000	C, D	K
G	12 days	$7,200	D	K, L, M
	5 days	$5,000	D	I
	3 days	$12,000	H	M
J	2 days	$1,600	E	N, R
K	5 days	$6,000	F, G	R
L	1 day	$1,500	G	O
M	2 days	$4,000	G, I	P, R
N	2 days	$8,000	J	Q
O	2 days	$1,200	L	R
P	4 days	$5,200	M	S
Q	2 days	$6,000	N	T
R	7 days	$7,000	J, K, M, O	V
S	2 days	$2,800	P	U
T	1 day	$5,000	Q	V
U	3 days	$3,000	S	V
V	5 days	$10,000	R, T, U	None

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Data for development of a CPM diagram for a project is shown below. Perform a **cost analysis** for the project similar to **Example 10-1** in your book. Develop the **CPM diagram** and show the **Early/Late Starts** and **Finishes**. Develop a **chart** that shows the **tabular form** of **Early/Late Starts** and **Finishes** and the **Total and Free Floats**. Develop a **tabular form** showing the **daily distribution of costs** on an **early start, late start,** and **target schedule**. **Plot** the **early start, late start,** and **target S-curves** on one graph to show the **results of your analysis**.

下方显示了用于开发项目的 CPM 图的数据。 请根据书中**示例 10-1** 对项目进行**成本分析**。绘制 **CPM** 图，并标注**最早/最晚开始时间**和**最早/最晚完成时间**。制作一个**表格**，显示**最早/最晚开始时间**和**最早/最晚完成时间**的**表格形式**，以及**总浮动时间**和**自由浮动时间**。制作一个**表格**，展示**按天分配的成本**，分别基于**最早开始、最晚开始**和**目标进度**。绘制**最早开始、最晚开始**和**目标 S 曲线**在同一张图上，以展示**分析结果**。

Activity	Duration	Cost	Preceded by	Followed by
A	2 days	$1,200	None	B, C, D
B	3 days	$3,300	A	E
C	8 days	$12,000	A	F
D	6 days	$18,000	A	F, G, H
E	8 days	$8,000	B	J
F	9 days	$27,000	C, D	K
G	12 days	$7,200	D	K, L, M
H	5 days	$5,000	D	I
I	3 days	$12,000	H	M
J	2 days	$1,600	E	N, R
K	5 days	$6,000	F, G	R
L	1 day	$1,500	G	O
M	2 days	$4,000	G, I	P, R
N	2 days	$8,000	J	Q
O	2 days	$1,200	L	R
P	4 days	$5,200	M	S
Q	2 days	$6,000	N	T
R	7 days	$7,000	J, K, M, O	V
S	2 days	$2,800	P	U
T	1 day	$5,000	Q	V
U	3 days	$3,000	S	V
V	5 days	$10,000	R, T, U	None

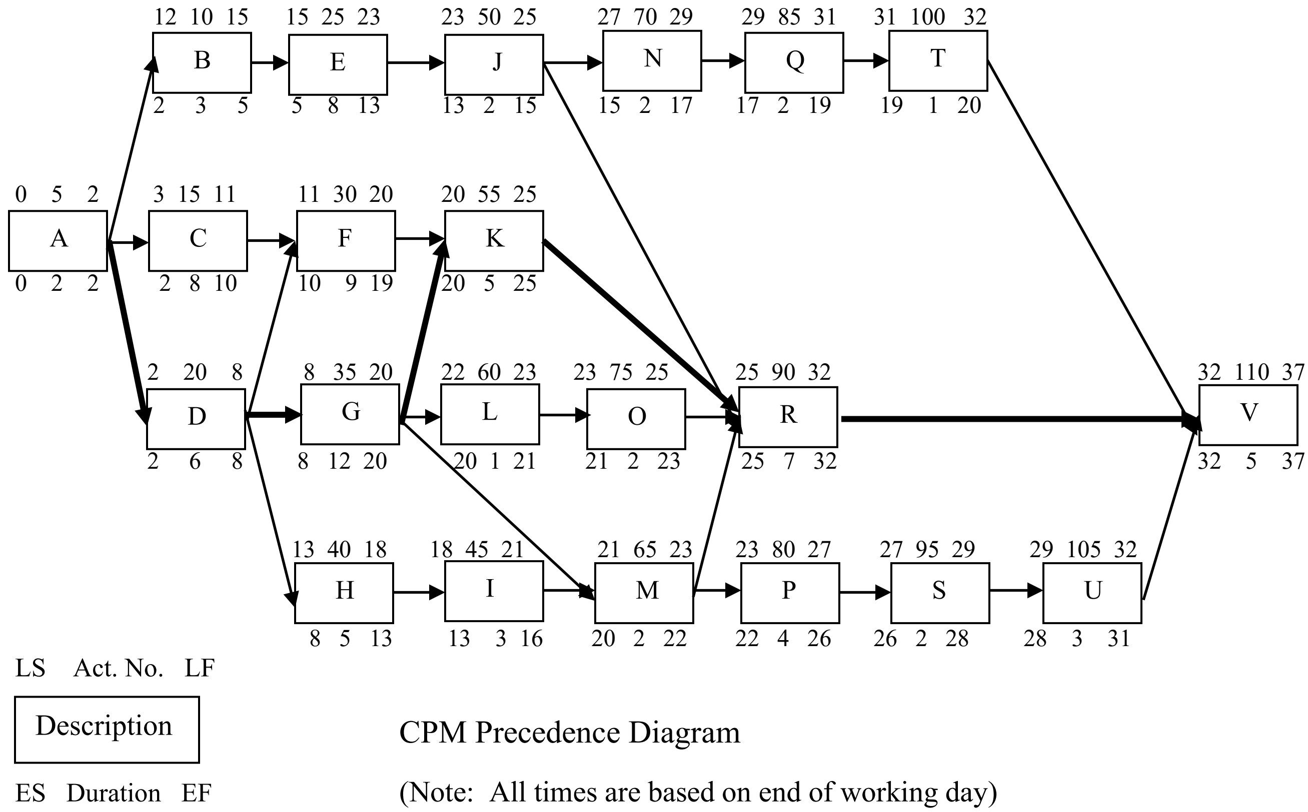
Chapter 10 - Project Scheduling

Question 6. **Data for development of a CPM diagram for a project is shown below. Perform a cost analysis for the project similar to Example 10-1 in your book. Develop the CPM diagram and show the Early/Late Starts and Finishes. Develop a chart that shows the tabular form of Early/Late Starts and Finishes and the Total and Free Floats. Develop a tabular form showing the daily distribution of costs on an early start, late start, and target schedule. Plot the early start, late start, and target S-curves on one graph to show the results of your analysis.**

Activity	Duration	Cost	Preceded by	Followed by
A	2 days	$1,200	None	B, C, D
B	3 days	$3,300	A	E
C	8 days	$12,000	A	F
D	6 days	$18,000	A	F, G, H
E	8 days	$8,000	B	J
F	9 days	$27,000	C, D	K
G	12 days	$7,200	D	K, L, M
H	5 days	$5,000	D	I
I	3 days	$12,000	H	M
J	2 days	$1,600	E	N, R
K	5 days	$6,000	F, G	R
L	1 day	$1,500	G	O
M	2 days	$4,000	G, I	P, R
N	2 days	$8,000	J	Q
O	2 days	$1,200	L	R
P	4 days	$5,200	M	S
Q	2 days	$6,000	N	T
R	7 days	$7,000	J, K, M, O	V
S	2 days	$2,800	P	U
T	1 day	$5,000	Q	V
U	3 days	$3,000	S	V
V	5 days	$10,000	R, T, U	None

Chapter 10 - Project Scheduling

Question 6 continued -
The CPM diagram and tabular schedule is shown on the following page.



CPM Precedence Diagram

(Note: All times are based on end of working day)

No.	Activity	Preceded By	Followed By	D	Total Cost	Early Start	Early Finish	Late Start	Late Finish	Total Float	Free Float	
5	A	None	B,C,D	2	$1,200	0	2	0	2	0	0	- CP
10	B	A	E	3	$3,300	2	5	12	15	10	0	
15	C	A	F	8	$12,000	2	10	3	11	1	0	
20	D	A	F,G,H	6	$18,000	2	8	2	8	0	0	- CP
25	E	B	J	8	$8,000	5	13	15	23	10	0	
30	F	C,D	K	9	$27,000	10	19	11	20	1	1	
35	G	D	K,L,M	12	$7,200	8	20	8	20	0	0	- CP
40	H	D	I	5	$5,000	8	13	13	18	5	0	
45	I	H	M	3	$12,000	13	16	18	21	5	4	
50	J	E	N,R	2	$1,600	13	15	23	25	10	0	
55	K	F,G	R	5	$6,000	20	25	20	25	0	0	- CP
60	L	G	O	1	$1,500	20	21	22	23	2	0	
65	M	G,I	P,R	2	$4,000	20	22	21	23	1	0	
70	N	J	Q	2	$8,000	15	17	27	29	12	0	
75	O	L	R	2	$1,200	21	23	23	25	2	2	
80	P	M	S	4	$5,200	22	26	23	27	1	0	
85	Q	N	T	2	$6,000	17	19	29	31	12	0	
90	R	J,K,M,O	V	7	$7,000	25	32	25	32	0	0	- CP
95	S	P	U	2	$2,800	26	28	27	29	1	0	
100	T	Q	V	1	$5,000	19	20	31	32	12	12	
105	U	S	V	3	$3,000	28	31	29	32	1	1	
110	V	R,T,U	None	5	$10,000	32	37	32	37	0	0	- CP

Chapter 10 - Project Scheduling

Question 6 continued -

Daily Distribution of Costs on an Early Start Basis:

Day	ES %-Time	Early Start Activities in Progress	ES Cost/Day	Cumulative	ES %-Cost
1	2.70%	A@$600	$600	$600	0.39%
2	5.41%	"	"	$1,200	0.77%
3	8.11%	B@$1,100 + C@$1,500 + D@$3,000	$5,600	$6,800	4.39%
4	10.81%	" " "	"	$12,400	8.00%
5	13.51%	" " "	"	$18,000	11.61%
6	16.22%	C@$1,500 + D@$3,000 + E@$1,000	$5,500	$23,500	15.16%
7	18.92%	" " "	"	$29,000	18.71%
8	21.62%	" " "	"	$34,500	22.26%
9	24.32%	C@$1,500 + E@ $1,000 + G@$600 + H@$1,000	$4,100	$38,600	24.90%
10	27.03%	" " " "	"	$42,700	27.55%
11	29.73%	E@$1,000 + F@$3,000 + G@$600 + H@$1,000	$5,600	$48,300	31.16%
12	32.42%	" " " "	"	$53,900	34.77%
13	35.14%	" " " "	"	$59,500	38.39%
14	37.84%	F@$3,000 + G@$600 + I@$4,000 + J@$800	$8,400	$67,900	43.81%
15	40.54%	" " " "	"	$76,300	49.23%
16	43.24%	F@$3,000 + G@$600 + I@$4,000 + N@$4,000	$11,600	$87,900	56.71%
17	45.95%	F@$3,000 + G@$600 + N@$4,000	$7,600	$95,500	61.61%
18	48.65%	F@$3,000 + G@$600 + Q@$3,000	$6,600	$102,100	65.87%
19	51.35%	" " "	"	$108,700	70.13%
20	54.05%	G@$600 + T@$5,000	$5,600	$114,300	73.74%
21	56.76%	K@$1,200 + L@$1,500 + M@$2,000	$4,700	$119,000	76.77%
22	59.46%	K@$1,200 + M@$2,000 + O@$600	$3,800	$122,800	79.23%
23	62.16%	K@$1,200 + O@$600 + P@$1,300	$3,100	$125,900	81.23%
24	64.86%	K@$1,200 + P@$1,300	$2,500	$128,400	82.84%
25	67.57%	" "	"	$130,900	84.45%
26	70.27%	P@$1,300 + R@$1,000	$2,300	$133,200	85.94%
27	72.97%	R@$1,000 + S@$1,400	$2,400	$135,600	87.48%
28	75.68%	" "	"	$138,000	89.03%
29	78.38%	R@$1,000 + U@$1,000	$2,000	$140,000	90.32%
30	81.08%	" "	"	$142,000	91.61%
31	83.78%	" "	"	$144,000	92.90%
32	86.49%	R@$1,000	$1,000	$145,000	93.55%
33	89.19%	V@$2,000	$2,000	$147,000	94.84%
34	91.89%	"	"	$149,000	96.13%
35	94.59%	"	"	$151,000	97.42%
36	97.30%	"	"	$153,000	98.71%
37	100.00%	"	"	$155,000	100.00%

Chapter 10 - Project Scheduling

Question 6 continued -

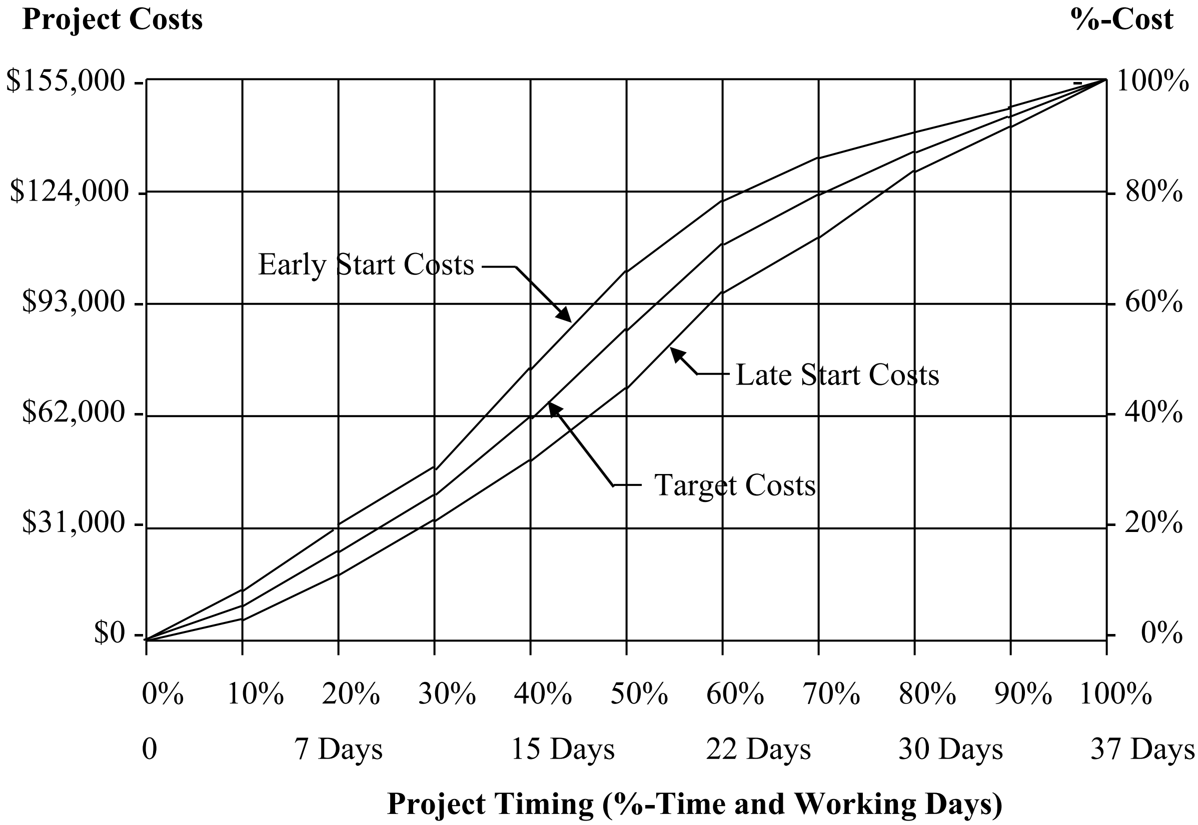
Daily Distribution of Costs on a Late Start Basis:

Day	LS %-Time	Late Start Activities in Progress	LS Cost/Day	Cumulative	LS %-Cost
1	2.70%	A@$600	$600	$600	0.39%
2	5.41%	“	“	$1,200	0.77%
3	8.11%	D@$3,000	$3,000	$4,200	2.71%
4	10.81%	C@$1,500 + D@$3,000	$4,500	$8,700	5.61%
5	13.51%	“ “	“	$13,200	8.52%
6	16.22%	“ “	“	$17,700	11.42%
7	18.92%	“ “	“	$22,200	14.32%
8	21.62%	“ “	“	$26,700	17.23%
9	24.32%	C@$1,500 + G@$600	$2,100	$28,800	18.58%
10	27.03%	“ “	“	$30,900	19.94%
11	29.73%	“ “	“	$33,000	21.29%
12	32.42%	F@$3,000 + G $600	$3,600	$36,600	23.61%
13	35.14%	B@$1,100 + F@$3,000 + G@$600	$4,700	$41,300	26.65%
14	37.84%	B@$1,100 + F@$3,000 + G@$600 + H@$1,000	$5,700	$47,000	30.32%
15	40.54%	“ “ “ "	“	$52,700	34.00%
16	43.24%	E@$1,000 + F@$3,000 + G@$600 + H@$1,000	$5,600	$58,300	37.61%
17	45.95%	“ “ “ “	“	$63,900	41.23%
18	48.65%	“ “ “ “	“	$69,500	44.84%
19	51.35%	E@$1,000 + F@$3,000 + G@$600 + I@$4,000	$8,600	$78,100	50.39%
20	54.05%	“ “ “ “	"	$86,700	55.94%
21	56.76%	E@$1,000 + I@$4,000 + K@$1,200	$6,200	$92,900	59.94%
22	59.46%	E@$1,000 + K@$1,200 + M@$2,000	$4,200	$97,100	62.65%
23	62.16%	E@$1,000 + K@$1,200 + L@$1,500 + M@$2,000	$5,700	$102,800	66.32%
24	64.86%	J@$800 + K@$1,200 + O@$600 + P@$1,300	$3,900	$106,700	68.84%
25	67.57%	“ “ “ “	“	$110,600	71.36%
26	70.27%	P@$1,300 + R@$1,000	$2,300	$112,900	72.86%
27	72.97%	“ “	“	$115,200	74.32%
28	75.68%	N@$4,000 + R@$1,000 + S@$1,400	$6,400	$121,600	78.45%
29	78.38%	“ “ “	“	$128,000	82.58%
30	81.08%	Q@$,3000 + R@$1,000 + U@$1,000	$5,000	$133,000	85.81%
31	83.78%	“ “ “	“	$138,000	89.03%
32	86.49%	R@$1,000+ T@$5,000 + U@$1,000	$7,000	$145,000	93.55%
33	89.19%	V@$2,000	$2,000	$147,000	94.84%
34	91.89%	“	“	$149,000	96.13%
35	94.59%	“	“	$151,000	97.42%
36	97.30%	“	“	$153,000	98.71%
37	100.00%	“	“	$155,000	100.00%

Chapter 10 - Project Scheduling

Question 6 continued –

S-Curve for the Project:



9. The **bar chart** shown on the next page is the **original baseline schedule** for a project that must be **completed in 25 working days**. The schedule was developed without regard to the **distribution of labor** on the project. The **number of crews** and the **number of workers** in each crew are shown for each activity.

下一页所示的**条形图**是项目的**初始基线进度表**，该项目必须在 **25 个工作日内**完成。该进度表的制定未考虑**劳动力分布**。每项活动所需的**班组数量**和**每个班组中的工人人数**也已列出。

Activity Number	Duration in Days	Number of Crews	Workers in Crew	Work Day of Project																								
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	3	1	4	X	X	X																						
B	7	2	3				X	X	X	X	X	X	X															
C	9	1	2					X	X	X	X	X	X	X	X	X												
D	7	2	5					X	X	X	X	X	X	X														
E	8	1	4							X	X	X	X	X	X	X	X											
F	10	3	4									X	X	X	X	X	X	X	X	X	X							
G	9	1	5										X	X	X	X	X	X	X	X	X							
H	8	3	6														X	X	X	X	X	X	X	X				
I	11	2	4														X	X	X	X	X	X	X	X	X	X	X	
J	4	3	2																				X	X	X	X		
K	6	1	3																				X	X	X	X	X	X

a. **Evaluate** the **allocation of labor** by preparing a **histogram** of the **distribution of labor**, similar to **Figure 8-13**.

a. 通过制作**直方图**评估**劳动力分配**，类似于**图 8-13** 所示。

b. Assume that the **completion date** of **25 working days** cannot be changed and that the **start of Activity A** and **finish of Activity K** cannot be changed. The start time for the remaining activities can be **delayed a maximum of 2 days**. **Reschedule** the project to **reduce any irregularity** in the distribution of labor, as illustrated in **Figures 8-13 and 8-14**.

b. 假设 **25 个工作日的完成日期**不能更改，且**活动 A 的开始时间**和**活动 K 的结束时间**也不能更改。其余活动的开始时间最多可**延迟 2 天**。重新**排程**项目，以**减少**劳动力分布中的**不规则性**，如**图 8-13 和图 8-14** 所示。

Activity Number	Duration in Days	Numbers of Crews	Workers in Crew	Work Day of Project																								
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	4	1	4	4	4	4	4																					
B	7	2	3			6	6	6	6	6	6	6																
C	9	1	2					2	2	2	2	2	2	2	2	2												
D	7	2	4					8	8	8	8	8	8	8														
E	3	1	5							5	5	5																
F	9	3	4									12	12	12	12	12	12	12	12	12								
G	4	1	5										5	5	5	5												
H	8	3	6														18	18	18	18	18	18	18	18				
I	11	2	4														8	8	8	8	8	8	8	8	8	8	8	
J	4	3	2																				6	6	6	6		
K	6	1	3																				3	3	3	3	3	3
			Totals	4	4	10	10	16	16	21	21	33	27	27	19	19	35	35	38	38	26	26	35	35	17	17	11	3

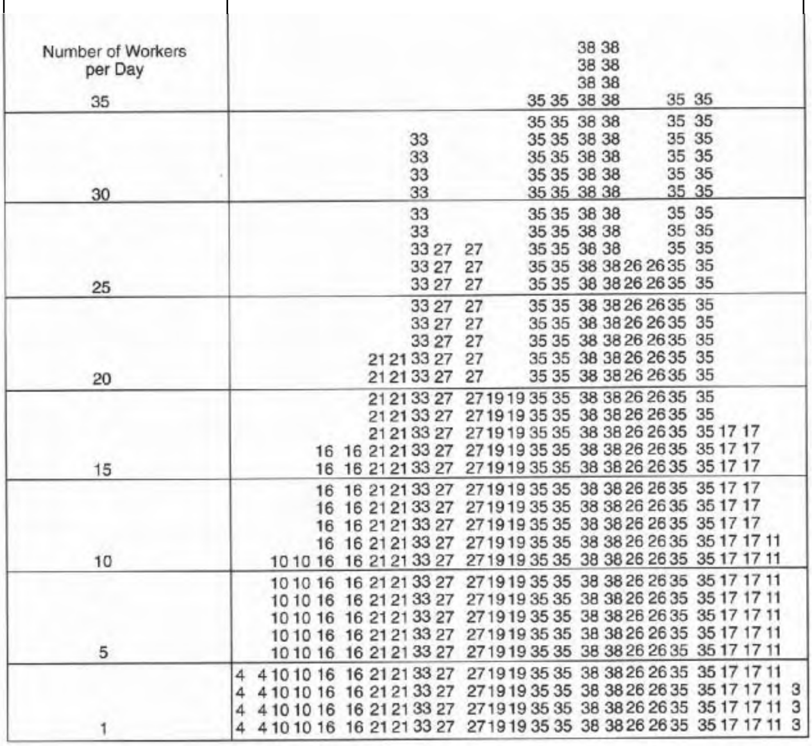


FIGURE 8-13
Irregular Distribution of Labor.

Activity Number	Duration in Days	Number of Crews	Workers in Crew	Work Day of Project																								
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	4	1	4	4	4	4	4																					
B	7	2	3			6	6	6	6	6	6	6																
C	9	1	2					2	2	2	2	2	2	2	2	2												
D	7	2	4					8	8	8	8	8	8	8														
E	3	1	5							5	5	5																
F	9	3	4										12	12	12	12	12	12	12	12	12							
G	4	1	5										5	5	5	5												
H	8	3	6												18	18	18	18	18	18	18	18						
I	11	2	4														8	8	8	8	8	8	8	8	8	8	8	
J	4	3	2																				6	6	6	6		
K	6	1	3																				3	3	3	3	3	3
			Totals	4	4	10	10	16	16	21	21	21	27	27	37	37	38	38	38	38	38	26	17	17	17	17	11	3

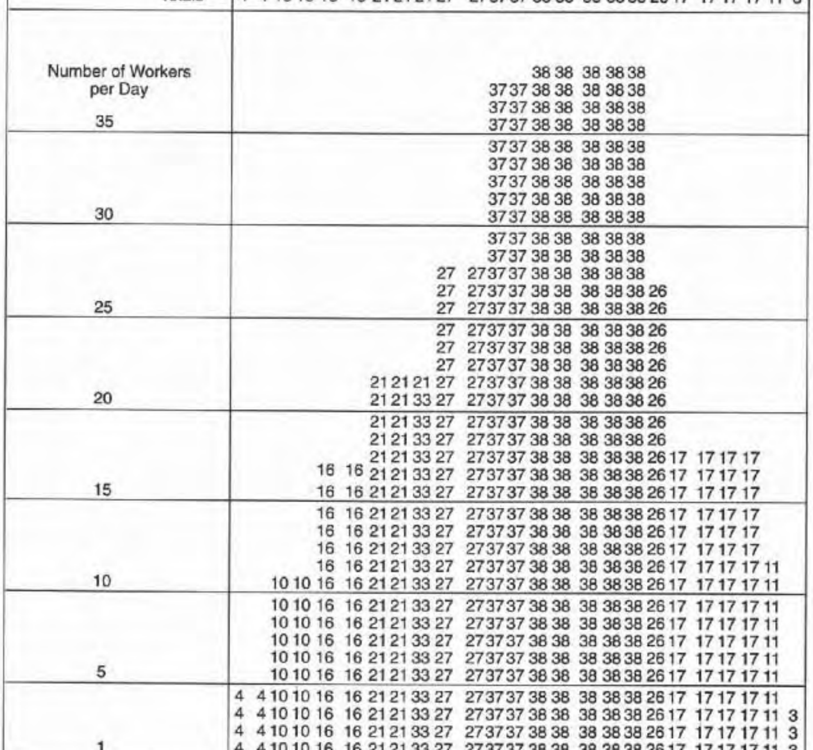


FIGURE 8-14
Distribution of Labor after Starting Activity F Two Days Later and Activity H Two Days Earlier.

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The bar chart shown below is the **original baseline schedule** for a project that must be **completed in 25 working days**. The schedule was developed **without regard to the distribution of labor** on the project. The **number of crews** and the **number of workers in each crew** are shown for each activity.

如下的条形图显示了**项目的原始基准进度**，该项目必须在 **25 个工作日内完成**。此进度表的制定**未考虑项目中的劳动分配**。每个活动的**工作组数量**和**每个工作组中的工人数**已显示。

Activity Number	Duration in Days	Workers of Crews	Workers in Crew	Work Day of Project				
				1 2 3 4 5	6 7 8 9 10	11 12131415	16 171819 20	21 22 23 24 25
A	3	1	4	XXX				
B	7	2	3	XX	X X X X X			
C	9	1	2	X	X X X X X	X X X		
D	7	2	5	X	X X X X X	X		
E	8	1	4		X X X X	X X X X		
F	10	3	4		X X	X X X X X	X X X	
G	9	1	5		X	X X X X X	X X X	
H	8	3	6			X X	X X X X X	X
I	11	2	4			X X	X X X X X	X X X X
J	4	3	2				X	X X X
K	6	1	3				X	X X X X X

(a) Evaluate the **allocation of labor** by preparing a **histogram of the distribution of labor**, similar to **Figure 10-14**. The **histogram is shown on Page 249**.

(a) 通过制作劳动分配的直方图，类似于**图 10-14**，**来评估劳动分配**。**直方图见第 249 页**。

(b) Assume that the **completion date of 25 working days** cannot be changed and that the **start of Activity A** and **finish of Activity K** cannot be changed. The **start time for the remaining activities** can be **delayed a maximum of 2 days**. **Reschedule** the project to **reduce any irregularity** in the **distribution of labor**, as illustrated in **Figures 10-14 and 10-15**. The **rescheduled histogram is shown on Page 249**.

(b) 假设 25 个工作日的完成日期不能更改，并且**活动 A 的开始时间**和**活动 K 的完成时间**不能更改。其余活动的**开始时间最多可以延迟 2 天**。**重新安排项目进度**以**减少劳动分配中的不规律性**，如**图 10-14 和图 10-15** 所示。**重新安排后的直方图见第 249 页**。

should include material and equipment required by the construction work force.

Construction labor represents a major cost of construction. The labor force on the job operates construction equipment and installs the materials. A resource allocation plan for construction is required to ensure high efficiency and productivity during construction. The project manager can resource load the project schedule to include the number of work-hours required for each craft of construction labor. The resource plan is a histogram of work-hours versus time, similar to the cost distribution analysis presented earlier in the chapter.

The construction plan shows the desired sequence of work. However, to be workable, the plan must also show the distribution of resources, such as the required labor for each craft on the job. The demand for labor should be uniformly distributed for each craft on the project, to prevent irregularities. The construction resource plan can be used as a tool to ensure a relatively uniform distribution of labor on the project. Resource loading the construction schedule also helps to identify and to prevent an over concentration of work in a particular area of the jobsite. Figure 10-14 is a simple bar chart for a project, showing each work activity, number of crews, and number of people in the crew for all labor in the project. The lower portion of Figure 10-14 shows irregularity in the distribution of workers per day during the middle of the project.

Activity Number	Duration in Days	Number of Crews	Workers in Crew	Work Day of Project 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	4	1	4	4	4	4	4																					
B	7	2	3			6	6	6	6	6	6	6																
C	9	1	2					2	2	2	2	2	2	2	2	2												
D	7	2	4					8	8	8	8	8	8	8														
E	3	1	5							5	5	5																
F	9	3	4									12	12	12	12	12	12	12	12	12								
G	4	1	5										5	5	5	5												
H	8	3	6														18	18	18	18	18	18	18	18				
I	11	2	4														8	8	8	8	8	8	8	8	8	8	8	
J	4	3	2																				6	6	6	6		
K	6	1	3																				3	3	3	3	3	3
			Totals	4	4	10	10	16	16	21	21	33	27	27	19	19	38	38	38	38	26	26	35	35	17	17	11	3

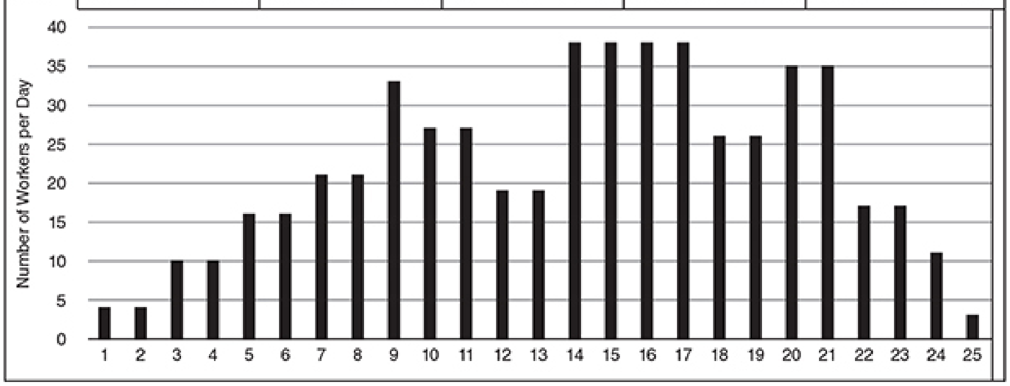


FIGURE 10-14 Irregular distribution of labor.

Figure 10-15 illustrates the same project as Figure 10-14, except Activity F is started 1 day later and Activity H is started 2 days earlier. The lower portion of Figure 10-15 shows a relative uniform distribution of labor by making these two minor adjustments in the project plan. A similar analysis can be made for each craft of labor to ensure uniform distributions of workers on the project. A resource allocation for a particular craft typically has a flat appearance on the graph, whereas a resource allocation for all crafts typically has a bell-shaped graph as shown in Figure 10-15.

Activity Number	Duration in Days	Number of Crews	Workers in Crew	Work Day of Project																								
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	4	1	4	4	4	4	4																					
B	7	2	3			6	6	6	6	6	6	6																
C	9	1	2					2	2	2	2	2	2	2	2	2												
D	7	2	4					8	8	8	8	8	8	8														
E	3	1	5							5	5	5																
F	9	3	4										12	12	12	12	12	12	12	12	12							
G	4	1	5										5	5	5	5												
H	8	3	6												18	18	18	18	18	18	18	18						
I	11	2	4														8	8	8	8	8	8	8	8	8	8	8	
J	4	3	2																				6	6	6	6		
K	6	1	3																				3	3	3	3	3	3
			Totals	4	4	10	10	16	16	21	21	21	27	27	37	37	38	38	38	38	38	26	17	17	17	17	11	3

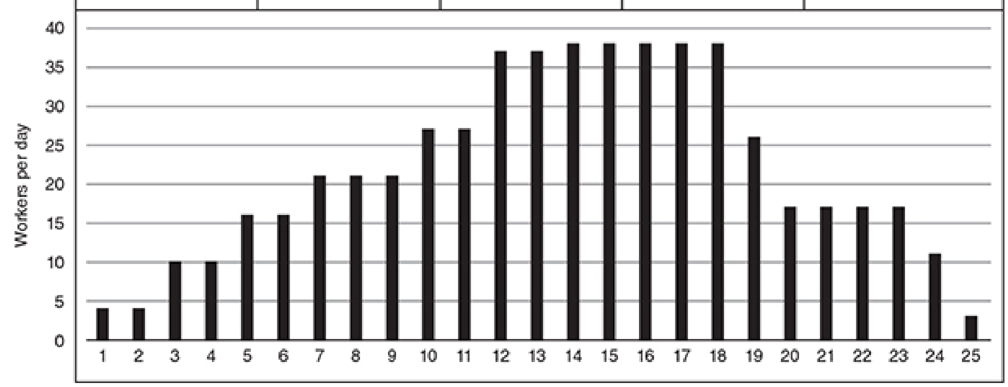


FIGURE 10-15 Labor distribution after starting Activity F 1 day later and Activity H 2 days earlier.

Calculations to Verify Schedules and Cost Distributions

Previous sections of this chapter presented the principles of planning, techniques of developing schedules, and methods of cost-loading CPM schedules to obtain the distribution of costs. Planning identifies the activities and sequencing of activities that must be performed to complete the project. Scheduling establishes the start and finish times of activities based on durations and interdependency of activities. The distribution of costs over the life of a project is obtained by assigning costs to activities from the cost estimate of the project.

The cost-loaded project schedule is crucial to managing a project because it connects the three fundamental components of a project: what needs to be done (scope), when will it be done (time), and how much it will cost (cost). Thus, a good-quality schedule and cost estimate are essential to

Chapter 10 - Project Scheduling

Question 7. **The bar char shown below is the original baseline schedule for a project that must be completed in 25 working days. The schedule was developed without regard to the distribution of labor on the project. The number of crews and the number of workers in each crew are shown for each activity.**

Activity Number	Duration in Days	Number of Crews	Workers in Cre w	Work Day of Project																								
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	3	1	4	X	X	X																						
B	7	2	3				X	X	X	X	X	X	X															
C	9	1	2					X	X	X	X	X	X	X	X	X												
D	7	2	5					X	X	X	X	X	X	X														
E	8	1	4							X	X	X	X	X	X	X	X											
F	10	3	4									X	X	X	X	X	X	X	X	X	X							
G	9	1	5										X	X	X	X	X	X	X	X	X							
H	8	3	6														X	X	X	X	X	X	X	X				
I	11	2	4														X	X	X	X	X	X	X	X	X	X	X	
J	4	3	2																				X	X	X	X		
K	6	1	3																				X	X	X	X	X	X

A.) Evaluate the allocation of labor by preparing a histogram of the distribution of labor, similar to Figure 10-14. The histogram is shown on Page 249.

B.) Assume that the completion date of 25 working days cannot be changed and that the start of Activity A and the finish of Activity K cannot be changed. The start time for the remaining activities can be delayed a maximum of 2 days. Reschedule the project to reduce any irregularity in the distribution of labor, as illustrated in Figures 10-14 and 10-15. The rescheduled histogram is shown on Page 249.

Question 7 continued –

Activity Number	Duration in Days	Number of Crews	Workers in Cre w	Work Day of Project																								
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	3	1	4	4	4	4																						
B	7	2	3				6	6	6	6	6	6	6															
C	9	1	2					2	2	2	2	2	2	2	2	2												
D	7	2	5					10	10	10	10	10	10	10														
E	8	1	4							4	4	4	4	4	4	4	4											
F	10	3	4									12	12	12	12	12	12	12	12	12	12							
G	9	1	5										5	5	5	5	5	5	5	5	5							
H	8	3	6														18	18	18	18	18	18	18	18				
I	11	2	4														8	8	8	8	8	8	8	8	8	8	8	
J	4	3	2																				6	6	6	6		
K	6	1	3																				3	3	3	3	3	3
			Totals	4	4	4	6	18	18	22	22	34	39	33	23	23	47	43	43	43	43	26	35	35	17	17	11	3

Number of Workers														47											
per day														47											
45														47											
														47											
														47	43	43	43								
														47	43	43	43								
														47	43	43	43								
40														47	43	43	43								
										39				47	43	43	43								
										39				47	43	43	43								
										39				47	43	43	43								
										39				47	43	43	43								
35										39				47	43	43	43			35	35				
									34	39				47	43	43	43			35	35				
									34	39	33			47	43	43	43			35	35				
									34	39	33			47	43	43	43			35	35				
									34	39	33			47	43	43	43			35	35				
30									34	39	33			47	43	43	43			35	35				
									34	39	33			47	43	43	43			35	35				
									34	39	33			47	43	43	43			35	35				
									34	39	33			47	43	43	43			35	35				
									34	39	33			47	43	43	43	26	26	35	35				
25									34	39	33			47	43	43	43	26	26	35	35				
									34	39	33			47	43	43	43	26	26	35	35				
									34	39	33	23	23	47	43	43	43	26	26	35	35				
							22	22	34	39	33	23	23	47	43	43	43	26	26	35	35				
							22	22	34	39	33	23	23	47	43	43	43	26	26	35	35				
20							22	22	34	39	33	23	23	47	43	43	43	26	26	35	35				
							22	22	34	39	33	23	23	47	43	43	43	26	26	35	35				
					18	18	22	22	34	39	33	23	23	47	43	43	43	26	26	35	35				
					18	18	22	22	34	39	33	23	23	47	43	43	43	26	26	35	35	17	17		
					18	18	22	22	34	39	33	23	23	47	43	43	43	26	26	35	35	17	17		
15					18	18	22	22	34	39	33	23	23	47	43	43	43	26	26	35	35	17	17		
					18	16	22	22	34	39	33	23	23	47	43	43	43	26	26	35	35	17	17		
					18	18	22	22	34	39	33	23	23	47	43	43	43	26	26	35	35	17	17		
					18	18	22	22	34	39	33	23	23	47	43	43	43	26	26	35	35	17	17		
					18	18	22	22	34	39	33	23	23	47	43	43	43	26	26	35	35	17	17	11	
10					18	18	22	22	34	39	33	23	23	47	43	43	43	26	26	35	35	17	17	11	
					18	18	22	22	34	39	33	23	23	47	43	43	43	26	26	35	35	17	17	11	
					18	18	22	22	34	39	33	23	23	47	43	43	43	26	26	35	35	17	17	11	
					18	18	22	22	34	39	33	23	23	47	43	43	43	26	26	35	35	17	17	11	
				6	18	18	22	22	34	39	33	23	23	47	43	43	43	26	26	35	35	17	17	11	
5				6	18	18	22	22	34	39	33	23	23	47	43	43	43	26	26	35	35	17	17	11	
	4	4	4	6	18	18	22	22	34	39	33	23	23	47	43	43	43	26	26	35	35	17	17	11	
	4	4	4	6	18	18	22	22	34	39	33	23	23	47	43	43	43	26	26	35	35	17	17	11	3
	4	4	4	6	18	18	22	22	34	39	33	23	23	47	43	43	43	26	26	35	35	17	17	11	3
1	4	4	4	6	18	18	22	22	34	39	33	23	23	47	43	43	43	26	26	35	35	17	17	11	3

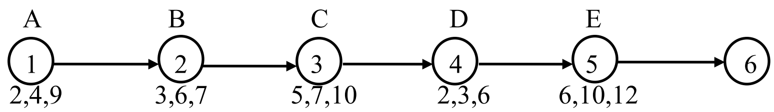
Question 7 continued –

Activity Number	Duration in Days	Number of Crews	Workers in Cre w	Work Day of Project																								
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	3	1	4	4	4	4																						
B	7	2	3				6	6	6	6	6	6	6															
C	9	1	2							2	2	2	2	2	2	2	2	2										
D	7	2	5							10	10	10	10	10	10	10												
E	8	1	4								4	4	4	4	4	4	4	4										
F	10	3	4											12	12	12	12	12	12	12	12	12	12					
G	9	1	5											5	5	5	5	5	5	5	5	5						
H	8	3	6																18	18	18	18	18	18	18	18		
I	11	2	4															8	8	8	8	8	8	8	8	8	8	8
J	4	3	2																					6	6	6	6	
K	6	1	3																				3	3	3	3	3	3
			Totals	4	4	4	6	6	6	18	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35	17	11

Number of Workers per day																									
45																									
																43	43	43	43						
																43	43	43	43						
																43	43	43	43	41					
40																43	43	43	43	41					
																43	43	43	43	41					
																43	43	43	43	41					
																43	43	43	43	41					
																43	43	43	43	41					
35																43	43	43	43	41	35	35	35		
																43	43	43	43	41	35	35	35		
											33	33	33			43	43	43	43	41	35	35	35		
											33	33	33			43	43	43	43	41	35	35	35		
											33	33	33		31	43	43	43	43	41	35	35	35		
30											33	33	33		31	43	43	43	43	41	35	35	35		
											33	33	33		31	43	43	43	43	41	35	35	35		
											33	33	33		31	43	43	43	43	41	35	35	35		
											33	33	33		31	43	43	43	43	41	35	35	35		
											33	33	33		31	43	43	43	43	41	35	35	35		
25											33	33	33		31	43	43	43	43	41	35	35	35		
											33	33	33		31	43	43	43	43	41	35	35	35		
											33	33	33	23	31	43	43	43	43	41	35	35	35		
							22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35		
							22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35		
20							22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35		
							22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35		
							22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35		
							22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35	17	
							22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35	17	
15							22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35	17	
							22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35	17	
							22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35	17	
							22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35	17	
							22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35	17	11
10							22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35	17	11
							22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35	17	11
						8	22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35	17	11
						8	22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35	17	11
				6	6	8	22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35	17	11
5				6	6	8	22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35	17	11
	4	4	4	6	6	8	22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35	17	11
	4	4	4	6	6	8	22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35	17	11
	4	4	4	6	6	8	22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35	17	11
1	4	4	4	6	6	8	22	22	22	22	33	33	33	23	31	43	43	43	43	41	35	35	35	17	11

Chapter 10 - Project Scheduling

Question 8. Below are the critical activities of a PERT diagram. The optimistic time (*a*), most likely time (*m*), and pessimistic time (*b*) are shown at the bottom of each activity. All durations are shown in weeks.



A.) Calculate the most likely time (t_e) for each activity. The expected time for each activity can be calculated by the following equation: $t_e = (a + 4m + b)/6$ where t_e is the expected time of an activity

Activity	$t_e = (a + 4m + b)/6$	Expected time - t_e
A	$[2 + 4(4) + 9]/6$	4.500
B	$[3 + 4(6) + 7]/6$	5.667
C	$[5 + 4(7) + 10]/6$	7.167
D	$[2 + 4(3) + 6]/6$	3.333
E	$[6 + 4(10) + 12]/6$	9.666

Summation = 30.333 weeks, which is T_E for Event 6.

B.) What is the optimistic, pessimistic, and expected time for completing the project?

The optimistic duration = 2 + 3 + 5 + 2 + 6 = 18
The pessimistic duration = 9 + 7 + 10 + 6 + 12 = 44
The expected duration = 4.50 + 5.667 + 7.167 + 3.333 + 9.666 = 30.333 weeks

C.) What is the standard deviation of the expected time for the last event, Event 6?

The σ_{TE} for the Event 6 is the square root of the variances of activities from the beginning of the project to the final event. σ_{TE} of Event 5 can be calculated as follows:

$$\sigma_{TE} = \sqrt{v_{1\text{-}2} + v_{2\text{-}3} + v_{3\text{-}4} + v_{4\text{-}5} + v_{5\text{-}6}}$$

$$= \sqrt{[(9\text{-}2)/6]^2 + [(7\text{-}3)/6]^2 + [(10\text{-}5)/6]^2 + [(6\text{-}2)/6]^2 + [(12\text{-}6)/6]^2}$$

$$= \sqrt{1.3611 + 0.4444 + 0.6944 + 0.4444 + 1.0000}$$

$$= 1.986$$

D.) What is the probability that this project can be completed in 32 weeks?

As calculated above, $T_E = 30.333$ and $\sigma_{TE} = 1.986$. The probability of completing the project by the 32nd week can be calculated as follows:

$$z = (T_S - T_E) / \sigma_{TE}$$
$$= (32 - 30.333) / 1.986$$
$$= 0.8394$$

From Table 10-8, based on a deviation z of 0.8394, the probability is 0.80. Thus, there is an 80% probability that the project will be completed by the 32nd week.

Chapter 10 - Project Scheduling

Question 8 continued -

E.) What is the probability that this project will be completed by the 28th week?

As calculated on the previous page, $\sigma_{TE} = 1.986$ and $T_E = 30.333$

$$\begin{aligned} z &= (T_S - T_E) / \sigma_{TE} \\ &= (28 - 30.333) / 1.986 \\ &= -1.175 \end{aligned}$$

From Table 10-8, based on a deviation z of -1.175, the probability is 0.13. Thus, there is a 13% probability that the project will be completed by the 28th week.

F.) For a 99% probability, what is expected number of days for completion.

As calculated on the previous page, $\sigma_{TE} = 1.986$ and $T_E = 30.333$
From Table 10-8 for a 99% probability, $z = +2.5$

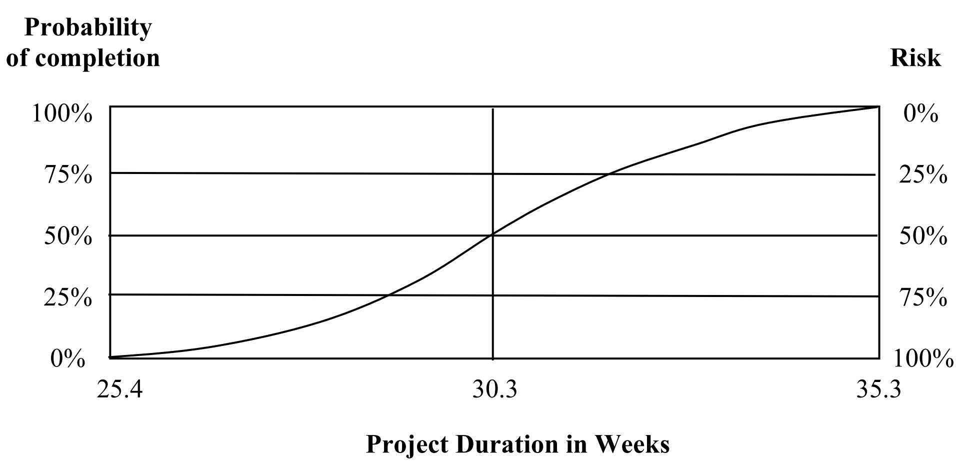
$$\begin{aligned} z &= (T_S - T_E) / \sigma_{TE} \\ +2.5 &= (T_S - 30.333) / 1.986 \\ T_S &= 35.3 \text{ weeks} \end{aligned}$$

G.) For a 1% probability, what is the expected number of days for completion?

As calculated on the previous page, $\sigma_{TE} = 1.986$ and $T_E = 30.333$
From Table 10-8 for a 1% probability, $z = -2.5$

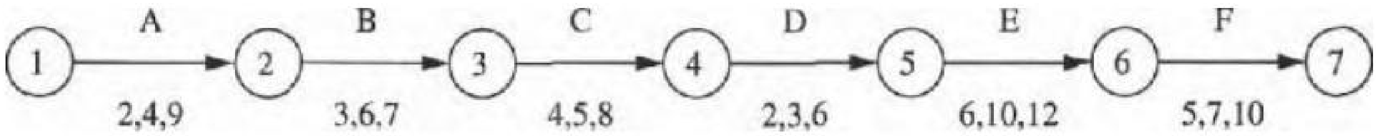
$$\begin{aligned} z &= (T_S - T_E) / \sigma_{TE} \\ -2.5 &= (T_S - 30.333) / 1.986 \\ T_S &= 25.4 \text{ weeks} \end{aligned}$$

H.) Develop a graphical plot of the probability versus the project duration for the full range of probabilities for assessing the risks of finishing the project ahead or behind the expected duration of 30.333 weeks.



10. Below are the **critical activities** of a **PERT diagram**. The **optimistic time (a)**, **most likely time (m)**, and **pessimistic time (b)** are shown at the bottom of each activity.

以下是一个 **PERT** 图表中的**关键活动**。每个活动的**乐观时间 (a)**、**最可能时间 (m)和悲观时间 (b)列在底部。**



a. **Calculate** the **most likely time (t_m)** for each activity.
b. What is the **optimistic**, **pessimistic**, and **expected time** for completing the project?
c. What is the **probability** that this project can be completed in **37 days**?
d. What is the probability that this project will be completed by the **thirty-second day**?
e. For a **99% probability**, what is the **expected number of days** for completion?
f. For a **1% probability**, what is the expected number of days for completion?

a. **计算每个活动的最可能时间 (t_m)。**
b. 项目的**乐观、悲观**和**预期完成时间**是多少？
c. 项目在 **37 天内完成**的**概率**是多少？
d. 项目在**第 32 天完成**的概率是多少？
e. 在 **99%的概率**下，项目预计需要**多少天**完成？
f. 在 **1%的概率**下，项目预计需要多少天完成？

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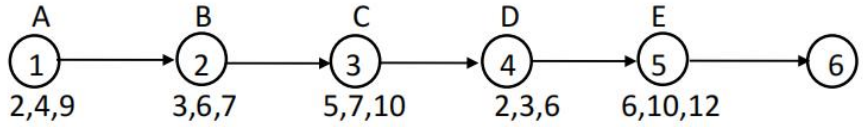
E4th Ch10 Q7

Below are the **critical activities** of a **PERT diagram.** The **optimistic time (a)**, **most likely time (m)**, and **pessimistic time (b)** are shown at the bottom of each activity. All durations are given in **weeks**.

以下是 PERT 图的关键活动。 每项活动的**乐观时间**（a）、**最可能时间**（m）和**悲观时间**（b）显示在各活动的底部，所有持续时间均以**周**为单位。



a. **Calculate the most likely time (t_e)** for each activity.
b. What is the **optimistic, pessimistic, and expected time** for completing the project?
c. What is the **standard deviation** of the expected time for the **last event, Event #6?**
d. What is the **probability** that this project can be completed in **32 weeks?**
e. What is the probability that this project will be completed **by the 28th week?**
f. For a **99% probability**, what is the **expected number of days** for completion?
g. For a **1% probability,** what is the **expected number of days** for completion?
h. **Develop a graphical plot of the probability versus the project duration** for the full range of probabilities to **assess the risks** of finishing the project **ahead of or behind** the expected duration of **30.333 weeks**.

a. 计算每项活动的最可能时间（te）。
b. 计算完成整个项目的乐观时间、悲观时间和期望时间。
c. 计算最后一个事件（事件 #6）期望时间的标准差。
d. 该项目在 32 周内完成的概率是多少？
e. 该项目在第 28 周前完成的概率是多少？
f. 如果要求 99%的完成概率，预计的完成天数是多少？
g. 如果只有 1%的完成概率，预计的完成天数是多少？
h. 绘制概率与项目持续时间的图形关系图，涵盖所有概率范围，以评估项目提前或滞后于期望持续时间 30.333 周 完成的风险。